



# **SJB INSTITUTE OF TECHNOLOGY**

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## **DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING**

### **DIGITAL SIGNAL PROCESSING (IPCC21EC42)**

### **MODULE-3 DESIGN OF FIR FILTERS**

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## MODULE-3 DESIGN OF FIR

### FILTERS

The impulse response of an Finite Impulse Response (FIR) filter is finite. The input-output relationship is

$$y(n) = \sum_{k=0}^m b_k x(n-k)$$

$b_k \rightarrow$  filter coefficients. Expanding and taking Z.T on both sides,

$$Y(z) = b_0 X(z) + b_1 z^{-1} X(z) + \dots + b_m z^{-m} X(z)$$

$$\therefore \text{The system function } H(z) = \frac{Y(z)}{X(z)}$$

$$H(z) = \frac{Y(z)}{X(z)} = b_0 + b_1 z^{-1} + \dots + b_m z^{-m}$$

FIR filter has no feedback in its equation. It makes the FIR filter inherently stable (no loops).

It can produce linear phase. An application requiring linear phase FIR must be used.

main disadvantage is time taken to execute.

Since there is no feedback, number of coeffs is more in the system equation to meet the

same system requirements as **IIR**. Hence complexity is more in terms of multiplications and memory requirements.

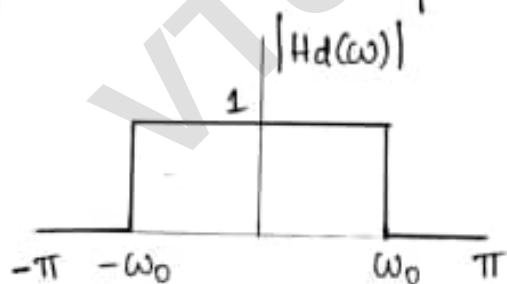
Let us consider a lowpass filter having frequency response given by

$$H_d(\omega) = \begin{cases} e^{-j\omega\alpha}, & |\omega| < \omega_0 \\ 0, & \omega_0 < |\omega| < \pi \end{cases} \quad (1)$$

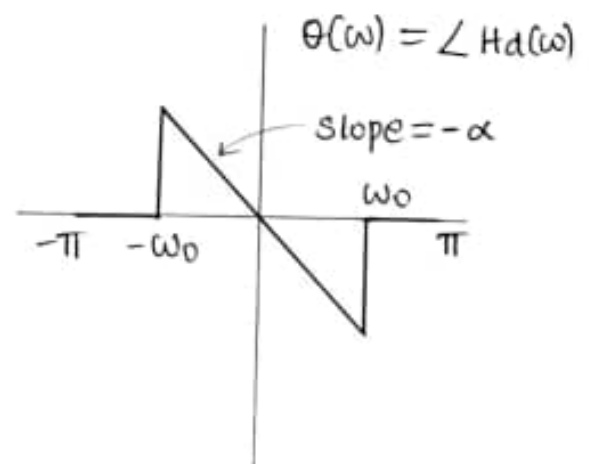
$H_d(\omega)$  is periodic with fundamental period  $2\pi$ .

ie,  $H_d(\omega + 2\pi m) = H_d(\omega)$ ,  $m \rightarrow \text{integer}$

The magnitude and phase frequency response of an ideal lowpass filter is as shown.



a) MAGNITUDE RESPONSE



b) PHASE RESPONSE

Let  $x(n)$  be the input and  $y(n)$  be the output of an ideal lowpass filter having an impulse response  $h_d(n)$ .

Since, lowpass filter is an LTI system, we have

$$y(n) = x(n) * h_d(n) \quad (2)$$

Taking DTFT on both sides,

$$Y(\omega) = X(\omega) H_d(\omega) \quad (3)$$

Substituting equation (1) in equation (3), we get

$$Y(\omega) = \begin{cases} X(\omega) e^{-j\omega\alpha}, & |\omega| < \omega_0 \\ 0, & \omega_0 < |\omega| < \pi \end{cases} \quad (4)$$

Taking inverse DTFT on both sides of Eqn (4),

$$y(n) = x(n - \alpha) \quad (5)$$

Hence, the linear phase has not altered the shape of  $x(n)$ , but simply delayed it by  $\alpha$ . Non-linear phase would have caused output to be distorted version of  $x(n)$ .

## SYMMETRIC AND ANTISYMMETRIC FIR FILTERS

A necessary and sufficient condition for linear-phase is that  $h(n)$  must be finite duration  $N$  and is either symmetric or antisymmetric about its midpoint.

SYMMETRIC :  $h(n) = h(N-1-n)$ ,  $n=0, 1, \dots, N-1$   
or

ANTISYMMETRIC :  $h(n) = -h(N-1-n)$ ,  $n=0, 1, \dots, N-1$

To prove that a linear-phase filter is obtained if we satisfy symmetry condition for  $h(n)$ .

Consider  $N$  is Even.

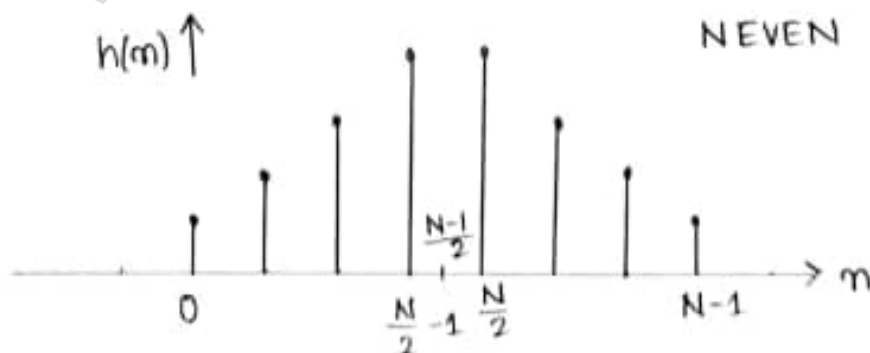


Fig. Impulse response  $h(n)$  that is symmetric with respect to midpoint for  $N$  even.

The z-transform of a causal sequence  $h(m)$  with  $m=0$  to  $N-1$  (finite) is given by

$$H(z) = \sum_{n=0}^{N-1} h(n) z^{-n} \quad \text{---(1)}$$

It can be broken into two summations as per the figure,

$$\therefore H(z) = \sum_{n=0}^{\frac{N}{2}-1} h(n) z^{-n} + \sum_{n=\frac{N}{2}}^{N-1} h(n) z^{-n} \quad \text{---(2)}$$

Let  $m = N-1-n$  in second summation,

$$\begin{aligned} H(z) &= \sum_{n=0}^{\frac{N}{2}-1} h(n) z^{-n} + \sum_{n=\frac{N}{2}}^{N-1} h(N-1-n) z^{-(N-1-n)} \\ &= \sum_{n=0}^{\frac{N}{2}-1} h(n) z^{-n} + \sum_{m=0}^{\frac{N}{2}-1} h(N-1-m) z^{-(N-1-m)} \quad \text{---(3)} \end{aligned}$$

Since  $m$  is a dummy variable, it can be replaced by  $n$ .

$$\begin{aligned} \text{Hence, } H(z) &= \sum_{n=0}^{\frac{N}{2}-1} h(n) z^{-n} + \sum_{n=0}^{\frac{N}{2}-1} h(N-1-n) z^{-(N-1-n)} \\ &\quad \text{---(4)} \end{aligned}$$



Since,  $h(N-1-m) = h(m)$  for  $m=0,1,\dots,N-1$   
(Symmetric)

$$\therefore H(z) = \sum_{m=0}^{\frac{N}{2}-1} h(m) [z^{-m} + z^{-(N-1-m)}]$$

To find the frequency response of the digital filter, letting  $z = e^{j\omega}$  in  $H(z)$ ,

$$\begin{aligned} H(e^{j\omega}) &= H(\omega) = \sum_{m=0}^{\frac{N}{2}-1} h(m) [e^{-j\omega m} + e^{-j\omega(N-1-m)}] \\ &= \sum_{m=0}^{\frac{N}{2}-1} 2 h(m) e^{-j\omega(\frac{N-1}{2})} \left[ \frac{e^{-j\omega[m - (\frac{N-1}{2})]} + e^{j\omega[m - (\frac{N-1}{2})]}}{2} \right] \end{aligned}$$

$$H(\omega) = e^{-j\omega(\frac{N-1}{2})} \sum_{m=0}^{\frac{N}{2}-1} 2 h(m) \cos\left(\omega\left[m - \left(\frac{N-1}{2}\right)\right]\right)$$

⑤

Comparing this with

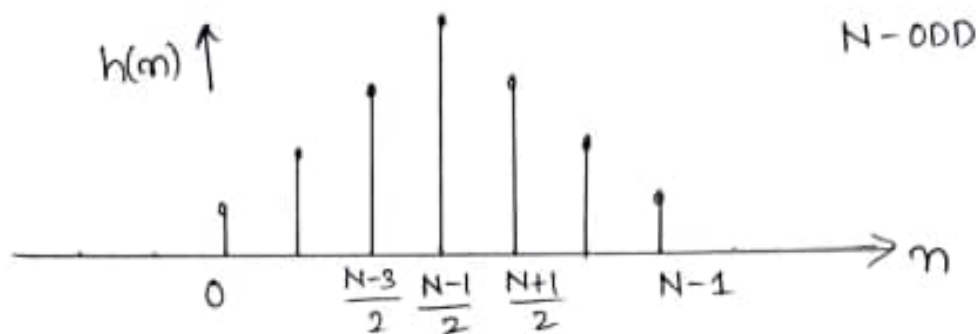
$$H(\omega) = H_r(\omega) e^{j\theta(\omega)} \quad H_r(\omega) \rightarrow \text{real function of } \omega.$$

$$H_r(\omega) = \sum_{m=0}^{\frac{N}{2}-1} 2 h(m) \cos\left(\omega\left[m - \left(\frac{N-1}{2}\right)\right]\right), \quad N \text{ Even}$$

$$\text{and } \theta(\omega) = \begin{cases} -\omega\left(\frac{N-1}{2}\right) & \text{if } H_r(\omega) > 0 \\ -\omega\left(\frac{N-1}{2}\right) + \pi & \text{if } H_r(\omega) < 0 \end{cases} \quad \text{⑥}$$

shows that  $\rightarrow$   
phase is linear

For N odd :



Since there is a value at the midpoint  $\frac{N-1}{2}$   
 for N-odd,  $H(z)$  <sup>from Equ ①</sup> could be expressed as,

$$H(z) = \sum_{n=0}^{\frac{(N-3)}{2}} h(n) z^{-n} + h\left(\frac{N-1}{2}\right) z^{-\left(\frac{N-1}{2}\right)} + \sum_{n=\frac{N+1}{2}}^{N-1} h(n) z^{-n} \quad \text{--- (7)}$$

Letting  $m = N-1-n$  in the third term on RHS,

$$H(z) = \sum_{n=0}^{\frac{(N-3)}{2}} h(n) z^{-n} + h\left(\frac{N-1}{2}\right) z^{-\left(\frac{N-1}{2}\right)} + \sum_{n=\frac{(N-3)}{2}}^0 h(N-1-n) z^{-(N-1-n)}$$

Since,  $n$  is a dummy variable, it can be replaced by  $n$ , and reversing order of summation

$$H(z) = \sum_{n=0}^{\frac{(N-3)}{2}} h(n) z^{-n} + h\left(\frac{N-1}{2}\right) z^{-\left(\frac{N-1}{2}\right)} + \sum_{n=0}^{\frac{(N-3)}{2}} h(N-1-n) z^{-(N-1-n)} \quad \text{--- (8)}$$



Making use of symmetry condition:  $h(N-1-m) = h(m)$ ,

$$H(z) = \sum_{n=0}^{\frac{(N-3)}{2}} h(n) \left[ z^{-n} + z^{-(N-1-n)} \right] + h\left(\frac{N-1}{2}\right) z^{-\left(\frac{N-1}{2}\right)} \quad \text{--- (9)}$$

Frequency response of digital filter is obtained by letting  $z = e^{j\omega}$  in  $H(z)$ .

$$\therefore H(e^{j\omega}) = H(\omega)$$

$$\begin{aligned} &= h\left(\frac{N-1}{2}\right) e^{-j\omega\left(\frac{N-1}{2}\right)} + \\ &\quad \sum_{n=0}^{(N-3)/2} h(n) \left[ e^{-j\omega n} + e^{-j\omega(N-1-n)} \right] \\ &= h\left(\frac{N-1}{2}\right) e^{-j\omega\left(\frac{N-1}{2}\right)} \\ &\quad + e^{-j\omega\left(\frac{N-1}{2}\right)} \sum_{n=0}^{\frac{(N-3)}{2}} 2h(n) \left( \frac{e^{-j\omega\left[n - \frac{(N-1)}{2}\right]} + e^{j\omega\left[n - \frac{(N-1)}{2}\right]}}{2} \right) \end{aligned}$$

$$H(\omega) = e^{-j\omega\left(\frac{N-1}{2}\right)} \left[ h\left(\frac{N-1}{2}\right) + \sum_{n=0}^{\frac{(N-3)}{2}} 2h(n) \cos\left(\omega\left[n - \frac{(N-1)}{2}\right]\right) \right]$$

L (10)

On comparing,

$$H_r(\omega) = h\left(\frac{N-1}{2}\right) + \sum_{n=0}^{\frac{N-3}{2}} 2 h(n) \cos\left(\omega\left[n - \left(\frac{N-1}{2}\right)\right]\right)$$

$$\text{and } \theta(\omega) = \begin{cases} -\omega\left(\frac{N-1}{2}\right) & \text{if } H_r(\omega) > 0 \\ -\omega\left(\frac{N-1}{2}\right) + \pi & \text{if } H_r(\omega) < 0 \end{cases}$$

$\theta(\omega)$  shows that it exhibits linear phase.

Similar expressions can be obtained for antisymmetric impulse response.

### Design of Linear-phase FIR filters using Windows

Ideal frequency response  $h_d(n)$  has infinite number of samples.

Truncating the impulse response of the ideal filter on both the right and the left sides thus yields a finite impulse response which approximates the behaviour of an ideal filter.

Abruptly truncating the impulse response results in sidelobes (ringing effect) as per "Gibbs phenomenon".

Hence a suitable window function is used to

minimize the effect of sidelobes,

$$h(m) = h_d(m) w(m)$$

multiplying by a suitable window function  $w(m)$  will make the infinite length  $h_d(m)$  into a finite impulse response  $h(m)$ .

Commonly used window types:

### 1) Rectangular window

The impulse response of an  $N$ -term (length) rectangular window is defined by

$$w_R(m) = \begin{cases} 1 & , 0 \leq m \leq N-1 \\ 0 & , \text{elsewhere} \end{cases}$$

The first sidelobe is 13 dB below the dc magnitude (0 dB).

The difference between passband and stopband gains is 21 dB for a low pass filter designed using this window.

### 2) Bartlett Window (or Triangular Window)

$$w_B(m) = \begin{cases} 1 - \frac{2 \left| m - \left( \frac{N-1}{2} \right) \right|}{N-1} & , 0 \leq m \leq N-1 \\ 0 & , \text{otherwise} \end{cases}$$

First sidelobe 27 dB below the dc magnitude.

Difference between PB and SB gain = 25 dB.

### 3) Hamming Window

$$w_{\text{Ham}}(n) = \begin{cases} 0.5 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right) & , 0 \leq n \leq N-1 \\ 0 & , \text{otherwise} \end{cases}$$

The smoother contour leads to smaller sidelobes than rectangular window.

First sidelobe 32 dB below the dc magnitude.

Difference between PB and SB gain = 44 dB

### 4) Hamming Window

$$w_{\text{Ham}}(n) = \begin{cases} 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right) & , 0 \leq n \leq N-1 \\ 0 & , \text{otherwise} \end{cases}$$

Smaller sidelobes due to smooth contour.

First sidelobe is 42 dB below the dc magnitude

Difference between PB and SB gain = 53 dB

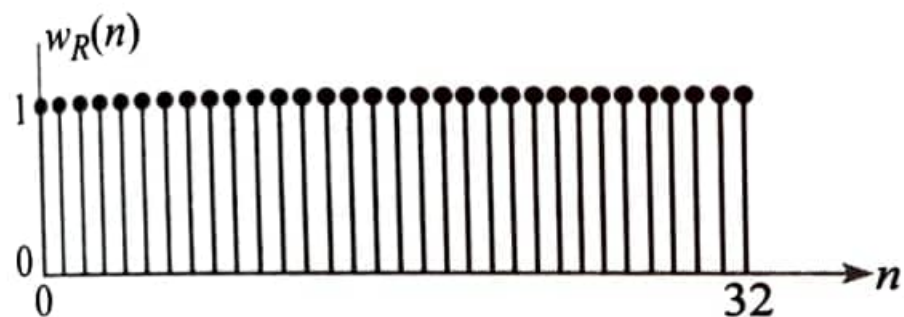


Fig. 5.7(a): Impulse response for rectangular window.

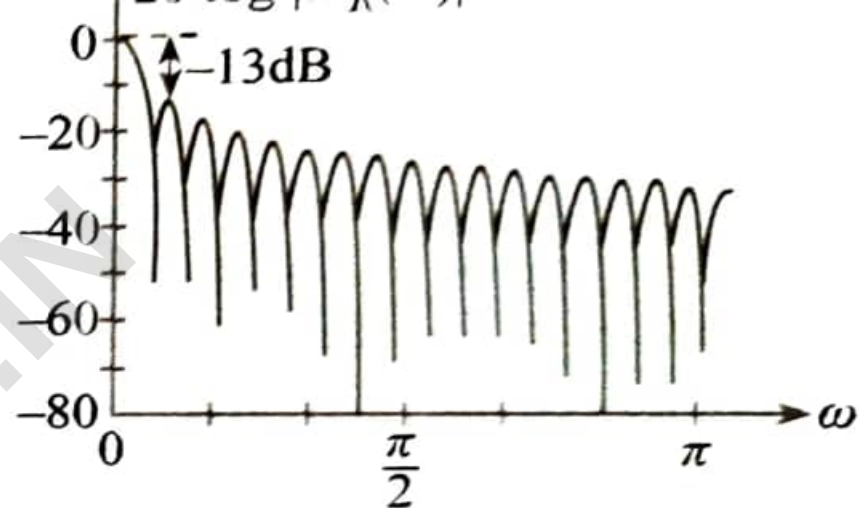


Fig. 5.7(b): Magnitude response for rectangular window.

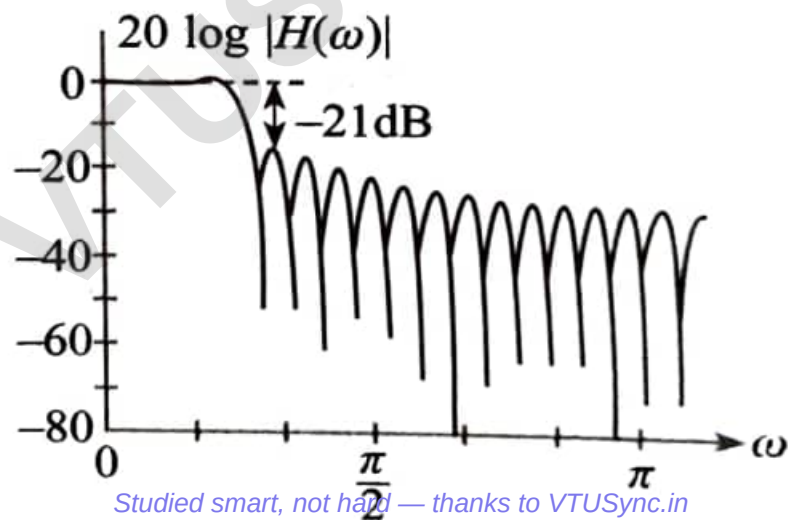
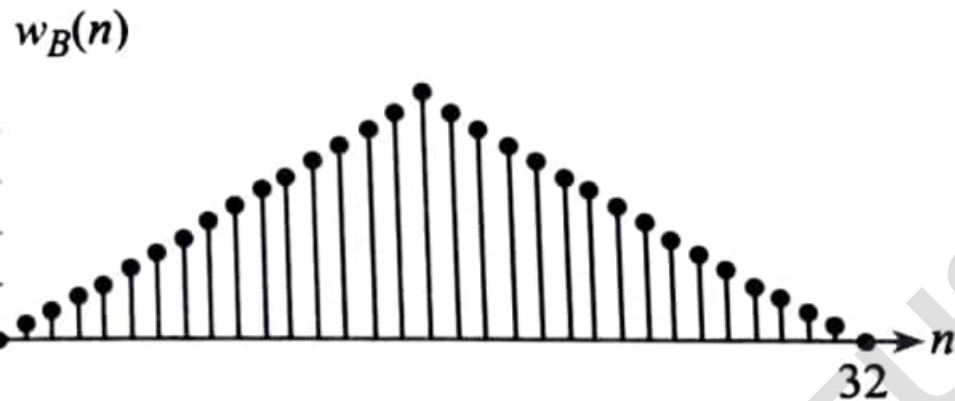
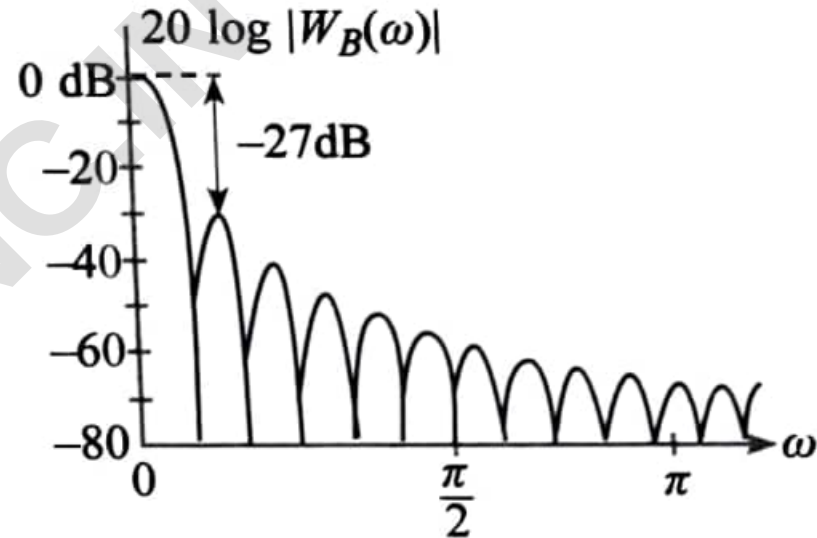


Fig. 5.7(c): Filter shape (magnitude response) for filter.

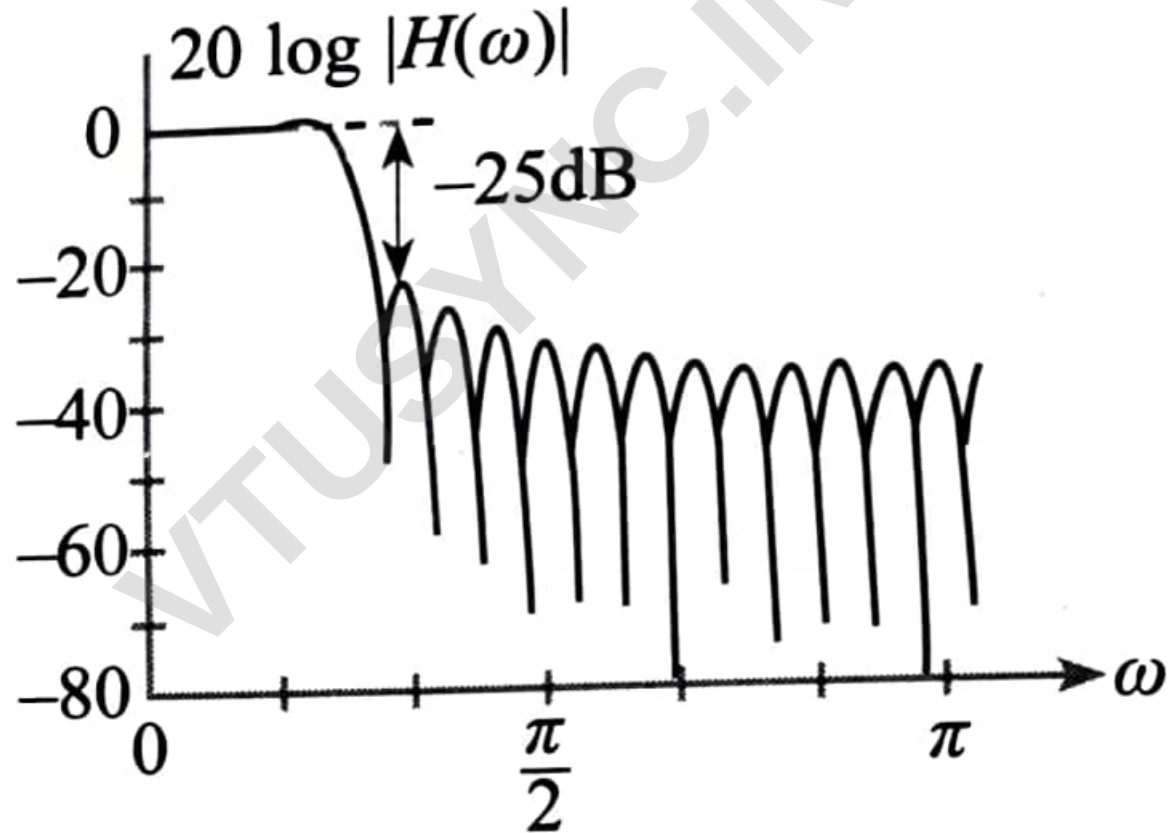


**Fig. 5.8 (a): Impulse response for Bartlett window**

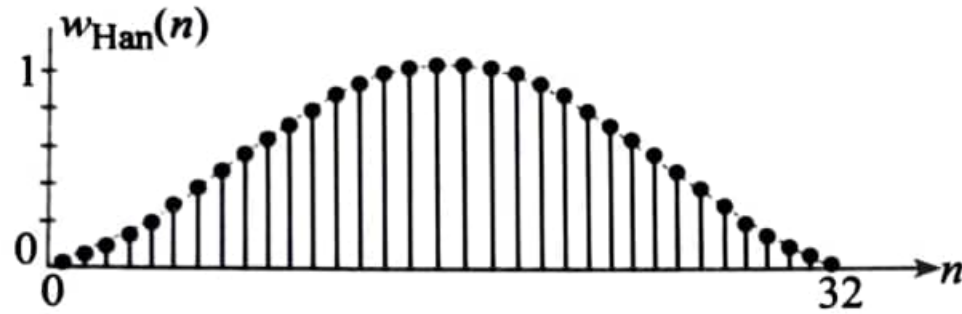


**Fig. 5.8 (b): Magnitude response for Bartlett window**

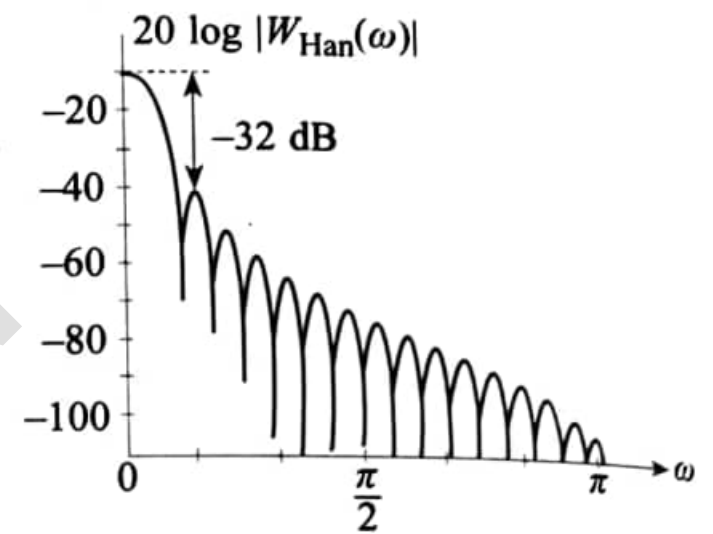




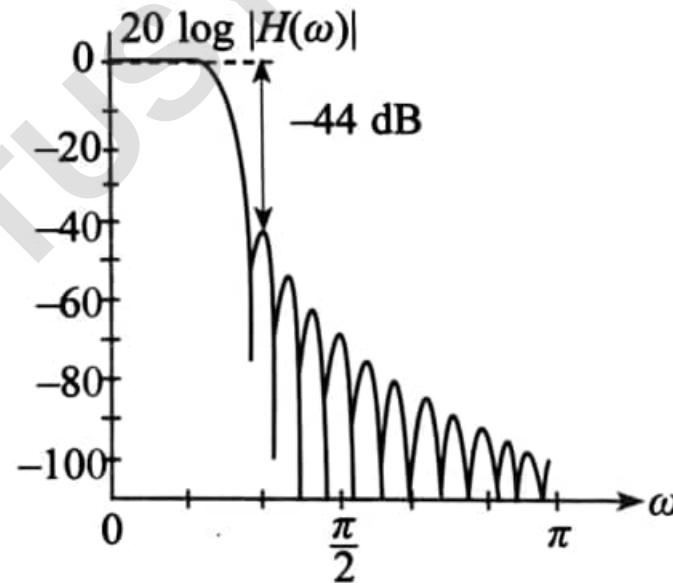
**Fig. 5.8(c): Filter shape for filter made with Bartlett window.**



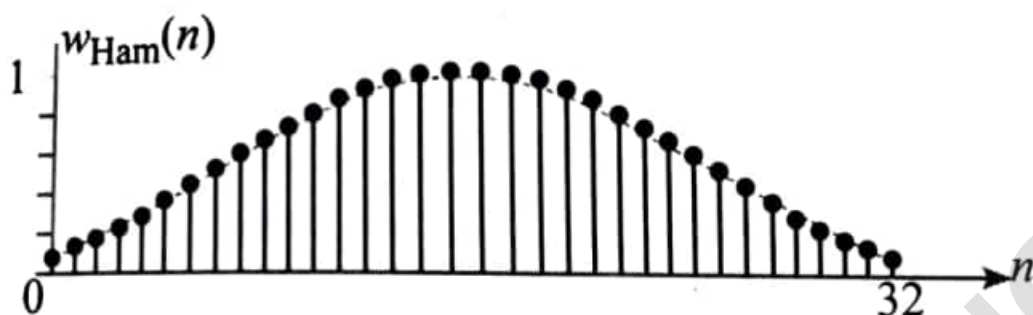
**Fig. 5.9(a): Impulse response for Hanning window**



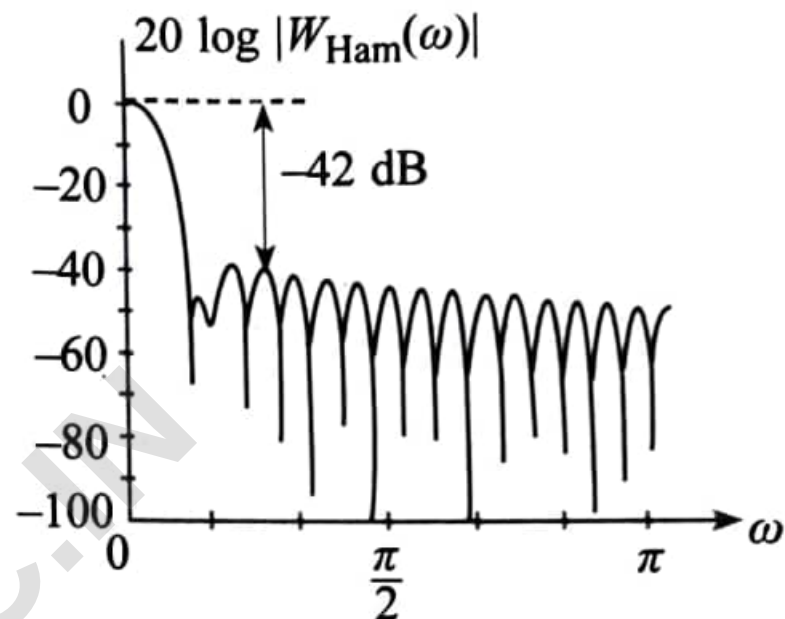
**Fig. 5.9(b): Magnitude response for Hanning window**



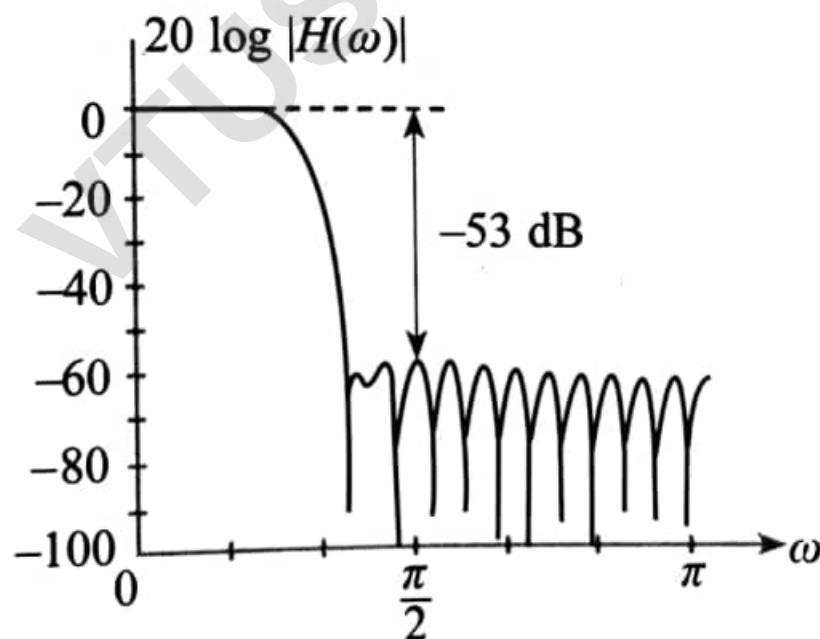
**Fig. 5.9(c): Filter shape for filter made with Hanning window.**



**Fig. 5.10(a): Impulse response for Hamming window**



**Fig. 5.10(b): Magnitude response for Hamming window**



**Fig. 5.10(c): Filter shape for filter made with Hamming window.**

## Design Procedure

An ideal low pass filter (desired filter) with a linear-phase of slope  $= -\alpha$  and a cutoff frequency  $\omega_c$  can be characterized by the following frequency response,

$$H_d(\omega) = \begin{cases} e^{-j\omega\alpha} & |\omega| < \omega_c \\ 0 & \omega_c < |\omega| < \pi \end{cases} \quad \text{--- (1)}$$

The impulse response  $h_d(n)$  is found by applying the Inverse DTFT,

hence,  $h_d(n) \triangleq \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega$

By defn -

$$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{-j\omega\alpha} e^{j\omega n} d\omega$$

From Eqn (1)

$$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega(n-\alpha)} d\omega$$

$$= \frac{1}{2\pi} \left[ \frac{e^{j\omega(n-\alpha)}}{j(n-\alpha)} \right]_{-\omega_c}^{\omega_c} = \frac{1}{2\pi j(n-\alpha)} \left[ e^{j\omega_c(n-\alpha)} - e^{-j\omega_c(n-\alpha)} \right]$$

$$h_d(n) = \frac{\sin[\omega_c(n-\alpha)]}{\pi(n-\alpha)}, \quad n \neq \alpha$$

Also, 
$$h_d(\alpha) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega\alpha} d\omega = \frac{1}{2\pi} [\omega]_{-\omega_c}^{\omega_c}$$

$$h_d(\alpha) = \frac{2\omega_c}{2\pi} = \frac{\omega_c}{\pi}, \quad \text{when } n=\alpha$$

$h_d(n)$  is noncausal in nature.

Multiplying with a causal and finite window with impulse response  $w(n)$ , yields  $h(n)$  which is causal.

$$h(n) = h_d(n) w(n)$$

$$h(n) = \begin{cases} \frac{\sin[\omega_c(n-\alpha)]}{\pi(n-\alpha)} w(n), & n \neq \alpha, 0 \leq n \leq N-1 \\ \frac{\omega_c}{\pi} w(n), & n = \alpha \end{cases}$$

\* For  $h(n)$  to be a linear phase filter  $\alpha$  must be chosen to be symmetric or antisymmetric about its midpoint.

$h_d(n)$  is symmetric about  $n = \alpha$  and

$w(n)$  is symmetric about  $n = \frac{N-1}{2}$

hence linear phase for  $h(n)$  results

when 
$$\alpha = \frac{N-1}{2}$$

| NAME OF THE WINDOW | TRANSITION WIDTH<br>$\Delta\omega = \omega_s - \omega_p$ | Stopband attenuation, dB<br>(minimum) | Window function<br>$w(m)$<br>$0 \leq m \leq N-1$  |
|--------------------|----------------------------------------------------------|---------------------------------------|---------------------------------------------------|
| Rectangular        | $\frac{4\pi}{N}$                                         | 21                                    | 1                                                 |
| Bartlett           | $\frac{8\pi}{N}$                                         | 25                                    | $1 - \frac{2 m - (\frac{N-1}{2}) }{N-1}$          |
| Hamming            | $\frac{8\pi}{N}$                                         | 44                                    | $0.5 - 0.5 \cos\left(\frac{2\pi m}{N-1}\right)$   |
| Hamming            | $\frac{8\pi}{N}$                                         | 53                                    | $0.54 - 0.46 \cos\left(\frac{2\pi m}{N-1}\right)$ |

Table: Summary of important features of window functions

Numerical 1: A lowpass filter is to be designed with the following desired frequency response

$$H_d(e^{j\omega}) = H_d(\omega) = \begin{cases} e^{-j2\omega} & , |\omega| \leq \frac{\pi}{4} \\ 0 & , \frac{\pi}{4} < |\omega| < \pi \end{cases}$$

Determine, the filter coeffs  $h_d(m)$  and  $h(m)$  if

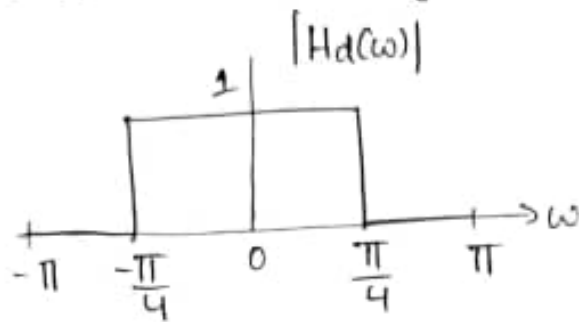
$w(m)$  is a rectangular window defined as

$$w_R(m) = \begin{cases} 1 & , 0 \leq m \leq 4 \\ 0 & , \text{otherwise} \end{cases}$$

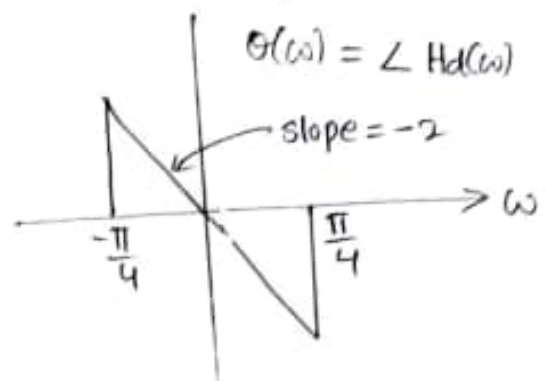
Also find frequency response  $H(\omega)$  of FIR filter.



Soln:- The magnitude and phase response,



Magnitude response  
of desired LPF



Phase response of the  
desired LPF

WKT, 
$$h_d(n) \triangleq \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega$$

Then, 
$$h_d(n) = \frac{1}{2\pi} \int_{-\pi/4}^{\pi/4} e^{-j2\omega} e^{j\omega n} d\omega$$

$$= \frac{\sin \left[ \frac{\pi}{4} (n-2) \right]}{\pi (n-2)}, \quad n \neq 2$$

and 
$$h_d(2) = \frac{1}{2\pi} \int_{-\pi/4}^{\pi/4} e^0 d\omega = \frac{\pi/4}{\pi} = \frac{1}{4}$$

Given, Rectangular window size is 5.

$$w_R(n) = \begin{cases} 1, & 0 \leq n \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

Then,  $h(m) = h_d(n) \omega_R(n)$

$$= \begin{cases} \sin\left[\frac{\pi}{4}(n-2)\right] \times 1 & , n=0,1,3,4 \\ \frac{1}{4} \times 1 & , n=2 \end{cases}$$

$n \neq 2$   
 $n = 2$

The coeff's are:

| $n$ | $h_d(m)$ | $\omega_R(m)$ | $h(m) = h_d(m) \omega_R(m)$ |
|-----|----------|---------------|-----------------------------|
| 0   | 0.159    | 1             | 0.159                       |
| 1   | 0.225    | 1             | 0.225                       |
| 2   | 0.25     | 1             | 0.25                        |
| 3   | 0.225    | 1             | 0.225                       |
| 4   | 0.159    | 1             | 0.159                       |

To find the frequency response, since  $N=5$  is odd.

$$\text{Hence } H(\omega) = e^{-j\omega\left(\frac{N-1}{2}\right)} \left[ h\left(\frac{N-1}{2}\right) + \sum_{n=0}^{\left(\frac{N-3}{2}\right)} 2 h(n) \cos\left[\omega\left(n - \left(\frac{N-1}{2}\right)\right)\right] \right]$$

with  $N=5$ ,

$$\begin{aligned} H(\omega) &= e^{-j2\omega} \left( h(2) + \sum_{n=0}^1 2 h(n) \cos[\omega(n-2)] \right) \\ &= e^{-j2\omega} \left( h(2) + 2 h(0) \cos 2\omega + 2 h(1) \cos \omega \right) \\ &= e^{-j2\omega} \left( 0.25 + 0.318 \cos 2\omega + 0.45 \cos \omega \right) \end{aligned}$$

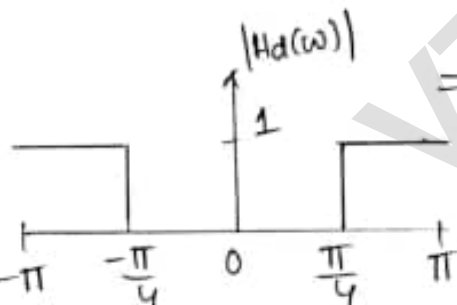
- ② A highpass FIR filter is to be designed with the following desired frequency response:

$$H_d(\omega) = \begin{cases} 0 & -\frac{\pi}{4} < \omega < \frac{\pi}{4} \\ e^{-j2\omega} & \frac{\pi}{4} < |\omega| < \pi \end{cases}$$

Find the frequency response of the FIR filter designed using rectangular window of length 5. (Five)

Soln:- The inverse DTFT of  $H_d(\omega)$  is

$$h_d(m) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega m} d\omega$$



$$= \frac{1}{2\pi} \int_{-\pi}^{-\pi/4} e^{-j2\omega} e^{j\omega m} d\omega + \frac{1}{2\pi} \int_{\pi/4}^{\pi} e^{-j2\omega} e^{j\omega m} d\omega$$

$$h_d(m) = \frac{1}{2\pi} \left[ \frac{e^{j\omega(m-2)}}{j(m-2)} \right]_{-\pi}^{-\pi/4} + \frac{1}{2\pi} \left[ \frac{e^{j\omega(m-2)}}{j(m-2)} \right]_{\pi/4}^{\pi}$$

$$= \frac{1}{2\pi j(m-2)} \left[ e^{-j\frac{\pi}{4}(m-2)} - e^{-j\pi(m-2)} \right] + \frac{1}{2\pi j(m-2)} \left[ e^{j\pi(m-2)} - e^{j\frac{\pi}{4}(m-2)} \right]$$

$$= \frac{1}{\pi(n-2)} \left[ \frac{e^{j\pi(n-2)} - e^{-j\pi(n-2)}}{2j} \right]$$

$$- \frac{1}{\pi(n-2)} \left[ \frac{e^{j\frac{\pi}{4}(n-2)} - e^{-j\frac{\pi}{4}(n-2)}}{2j} \right]$$

$$h_d(n) = \frac{1}{\pi(n-2)} \left\{ \sin \pi(n-2) - \sin \frac{\pi}{4}(n-2) \right\}, n \neq 2$$

$$\text{and } n=2, \quad h_d(2) = \frac{1}{2\pi} \int_{-\pi}^{-\frac{\pi}{4}} e^0 d\omega + \frac{1}{2\pi} \int_{\frac{\pi}{4}}^{\pi} e^0 d\omega$$

$$= \frac{1}{2\pi} \left[ \frac{3\pi}{4} + \frac{3\pi}{4} \right] = \underline{\underline{\frac{3}{4}}}$$

The impulse response of the highpass FIR filter

is,  $h(n) = h_d(n) w_R(n), \quad 0 \leq n \leq 4$

$$\text{where } w_R(n) = \begin{cases} 1, & 0 \leq n \leq 4 \\ 0, & \text{otherwise} \end{cases}$$

Coeff's are calculated,

| $n$ | $h_d(n)$ | $w_R(n)$ | $h(n) = h_d(n)w_R(n)$ |
|-----|----------|----------|-----------------------|
| 0   | -0.159   | 1        | -0.159                |
| 1   | -0.225   | 1        | -0.225                |
| 2   | 0.75     | 1        | 0.75                  |
| 3   | -0.225   | 1        | -0.225                |
| 4   | -0.159   | 1        | -0.159                |

Since  $N=5$  is odd, the frequency response of the FIR filter is,

$$H(e^{j\omega}) = H(\omega) = e^{-j\omega\left(\frac{N-1}{2}\right)} \left[ h\left(\frac{N-1}{2}\right) + \sum_{m=0}^{(N-3)/2} 2 h(m) \cos\left[\omega\left(m - \left(\frac{N-1}{2}\right)\right)\right] \right]$$

$$\begin{aligned} \text{For } N=5, \quad H(\omega) &= e^{-j2\omega} \left[ h(2) + 2h(0) \cos 2\omega + 2h(1) \cos \omega \right] \\ &= e^{-j2\omega} \left[ 0.75 - 2 \times 0.159 \cos 2\omega + 2 \times -0.225 \cos \omega \right] \\ &= e^{-j2\omega} \left[ 0.75 - 0.318 \cos 2\omega - 0.45 \cos \omega \right] \end{aligned}$$

- ③ The frequency response of an FIR filter is given by  $H(\omega) = e^{-j3\omega} (1 + 1.8 \cos 3\omega + 1.2 \cos 2\omega + 0.5 \cos \omega)$ . Determine the coefficients of the impulse response  $h(m)$  of the FIR filter.

Soln:- From the given frequency response,

$$\alpha = \frac{N-1}{2} = 3 \Rightarrow N=7$$

Since,  $N=7$  odd, the frequency response of the center symmetric FIR filter is given by

$$\begin{aligned}
 H(\omega) &= e^{-j\omega\left(\frac{N-1}{2}\right)} \left[ h\left(\frac{N-1}{2}\right) + \sum_{n=0}^{\left(\frac{N-3}{2}\right)} 2h(n) \cos\left[\omega\left(n - \left(\frac{N-1}{2}\right)\right)\right] \right] \\
 &= e^{-j3\omega} \left( h(3) + \sum_{n=0}^2 2h(n) \cos[\omega(n-3)] \right) \\
 &= e^{-j3\omega} \left[ h(3) + 2h(0) \cos 3\omega + 2h(1) \cos 2\omega + 2h(2) \cos \omega \right]
 \end{aligned}$$

Comparing with given equation in the data,

$$\begin{aligned}
 h(3) &= 1, \quad h(0) = 0.9, \quad h(1) = 0.6 \quad \text{and} \\
 h(2) &= 0.25
 \end{aligned}$$

The rest of the points could be found from symmetry,  $h(n) = h(N-1-n)$

$$h(n) = h(6-n), \quad n=0, 1, \dots, 6$$

$$h(6) = h(0) = 0.9$$

$$h(5) = h(1) = 0.6$$

$$h(4) = h(2) = 0.25$$

$$\text{Hence, } h(n) = \{ 0.9, 0.6, 0.25, 1, 0.25, 0.6, 0.9 \}$$



④ The desired frequency response of a lowpass FIR filter is given by

$$H_d(e^{j\omega}) = H_d(\omega) = \begin{cases} e^{-j3\omega}, & |\omega| < \frac{3\pi}{4} \\ 0, & \frac{3\pi}{4} < |\omega| < \pi \end{cases}$$

Determine the frequency response of the FIR filter, if Hamming window is used with  $N=7$ .

Soln:- By def'n of the inverse DTFT,

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\frac{3\pi}{4}}^{\frac{3\pi}{4}} e^{-j3\omega} e^{j\omega n} d\omega$$

$$= \frac{\sin\left[\frac{3\pi}{4}(n-3)\right]}{\pi(n-3)}, \quad n \neq 3$$

$$h_d(n) \text{ for } n=3, \quad h_d(3) = \frac{1}{2\pi} \int_{-\frac{3\pi}{4}}^{\frac{3\pi}{4}} e^0 d\omega = \frac{\frac{3\pi}{4}}{\pi}$$

$$h_d(3) = \frac{3}{4}$$

The impulse response of the lowpass FIR filter is

$$h(n) = h_d(n) w_{\text{Ham}}(n), \quad 0 \leq n \leq 6$$

$$\text{where } w_{\text{Ham}}(n) = \begin{cases} 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right) & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$$

for  $N=7$ ,

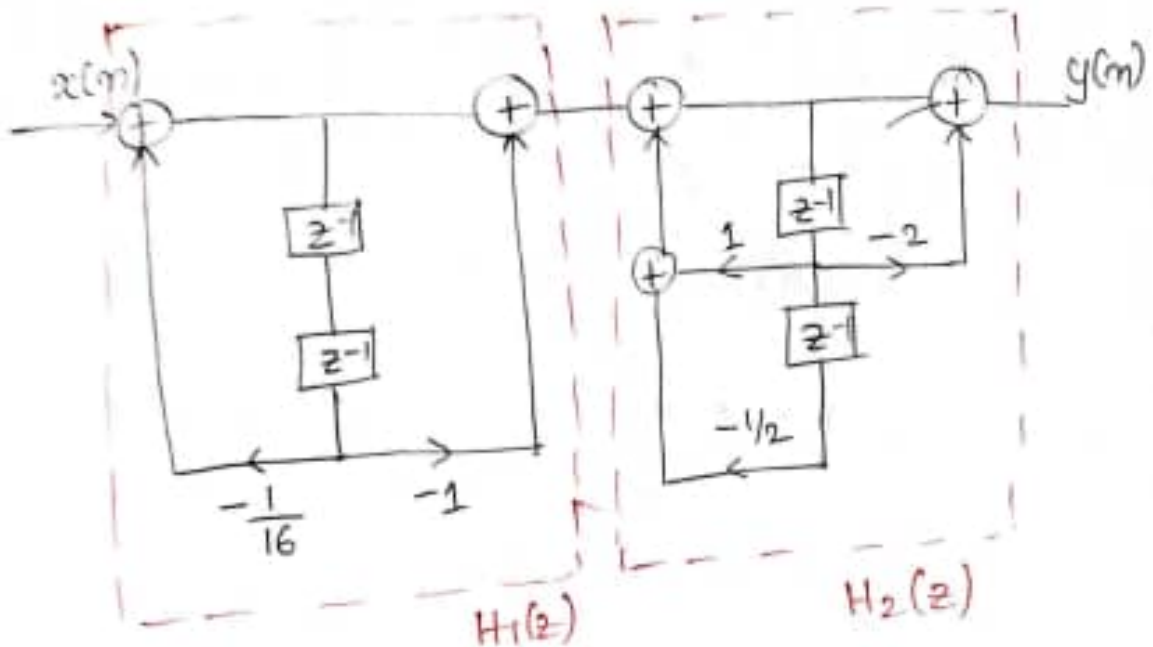
$$w_{\text{Ham}}(n) = \begin{cases} 0.54 - 0.46 \cos\left(\frac{2\pi n}{6}\right) & 0 \leq n \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

Hence, the coeff's  $h_d(n)$ ,  $w_{\text{Ham}}(n)$  &  $h(n)$  are obtained,

| $n$ | $h_d(n)$ | $w_{\text{Ham}}(n)$ | $h(n) = h_d(n) w_{\text{Ham}}(n)$ |
|-----|----------|---------------------|-----------------------------------|
| 0   | 0.075    | 0.08                | 0.006                             |
| 1   | -0.159   | 0.31                | -0.049                            |
| 2   | 0.225    | 0.77                | 0.173                             |
| 3   | 0.75     | 1                   | 0.75                              |
| 4   | 0.225    | 0.77                | 0.173                             |
| 5   | -0.159   | 0.31                | -0.049                            |
| 6   | 0.075    | 0.08                | 0.006                             |

Since  $N$  is odd, the frequency response of the FIR filter is,

$$\begin{aligned} H(e^{j\omega}) &= H(\omega) = e^{-j\omega\left(\frac{N-1}{2}\right)} \left[ h\left(\frac{N-1}{2}\right) + \sum_{n=0}^{\frac{N-3}{2}} 2h(n) \cos\left[\omega\left(n - \frac{N-1}{2}\right)\right] \right] \\ &= e^{-j3\omega} \left[ h(3) + 2h(0) \cos 3\omega + 2h(1) \cos 2\omega + 2h(2) \cos \omega \right] \\ &= e^{-j3\omega} \left[ 0.75 + 2 \times 0.006 \cos 3\omega + 2 \times -0.049 \cos 2\omega \right. \\ &\quad \left. + 2 \times 0.173 \cos \omega \right] \\ &= e^{-j3\omega} \left[ 0.75 + 0.012 \cos 3\omega - 0.098 \cos 2\omega \right. \\ &\quad \left. + 0.346 \cos \omega \right] \end{aligned}$$



(m-1) → MEM LOCATIONS  
 m → MULTIPLICATION  
 Obtain Direct Form Realization for TRANSVERSAL OR TAPPED-DELAY LINE FILTER

$$H(z) = 1 + 2z^{-1} - 3z^{-2} - 4z^{-3} + 5z^{-4}$$

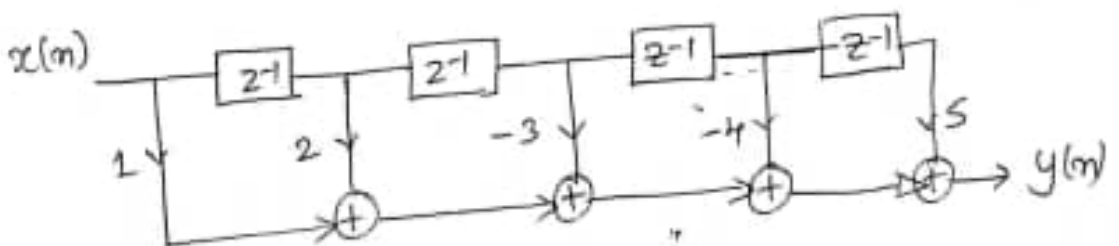
Soln:-

$$H(z) = \frac{y(z)}{x(z)} = 1 + 2z^{-1} - 3z^{-2} - 4z^{-3} + 5z^{-4}$$

$$\text{or } y(z) = x(z) + 2z^{-1}x(z) - 3z^{-2}x(z) - 4z^{-3}x(z) + 5z^{-4}x(z)$$

taking IZT,

$$y(m) = x(m) + 2x(m-1) - 3x(m-2) - 4x(m-3) + 5x(m-4)$$



DF-I REALIZATION

⑧ Obtain cascade realization

$$H(z) = (1 + 2z^{-1} - z^{-2})(1 + z^{-1} - z^{-2})$$

$$\text{Let } H(z) = H_1(z) \cdot H_2(z)$$

$$H_1(z) = \frac{y_1(z)}{x_1(z)} = 1 + 2z^{-1} - z^{-2}$$

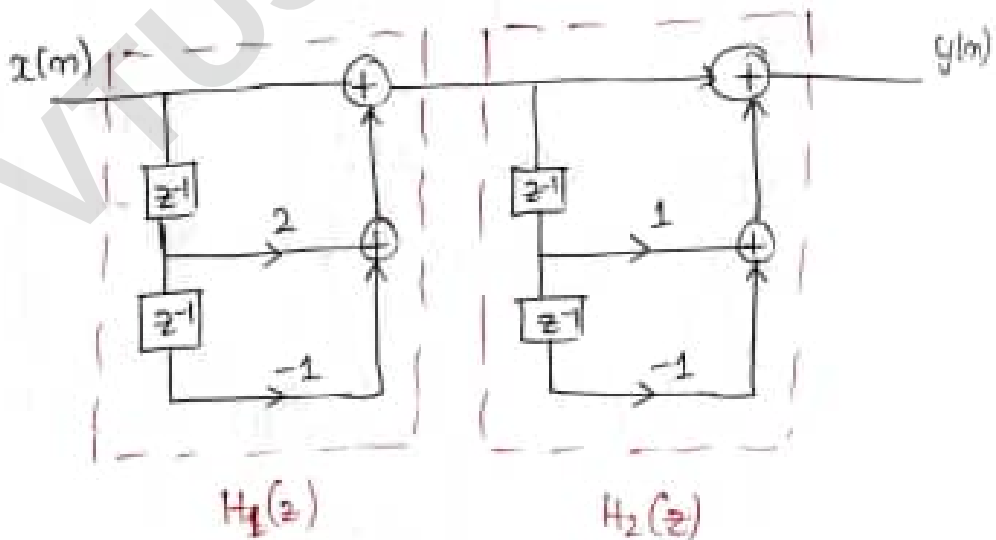
taking IZT & simplify.

$$y_1(m) = x_1(m) + 2x_1(m-1) - x_1(m-2)$$

Similarly

$$H_2(z) = \frac{y_2(z)}{x_2(z)} = 1 + z^{-1} - z^{-2}$$

$$y_2(m) = x_2(m) + x_2(m-1) - x_2(m-2)$$



(P) An FIR filter is given by

$$y(m) = x(m) + \frac{2}{5} x(m-1) + \frac{3}{4} x(m-2) + \frac{1}{3} x(m-3)$$

Sketch the lattice structure.

Lattice Structure of FIR filters:-

(P) Draw the lattice structure of FIR filter with system function

$$H(z) = 1 + 2z^{-1} + \frac{1}{3} z^{-2}$$

Soln:- From system function,

$$a_2(1) = 2, \quad a_2(2) = 1/3$$

↳ Second order

$M = 2$  (order / two zeros)

DF TO LATTICE CONVERSION  $m = m, m-1, \dots, 1$

W.K.T

$$k_m = a_m(m), \quad a_{m-1}(0) = 1$$

$$a_{m-1}(i) = \frac{a_m(i) - a_m(m) a_m(m-i)}{1 - k_m^2}$$

$1 \leq i \leq m-1$

With  $m = 2$ ,

$$k_2 = a_2(2) = 1/3$$

[ $k \rightarrow$  Reflection Coefficient]

$$m = 2, \quad k_1 = a_1(1)$$



$$m=2 \text{ \& } i=1$$

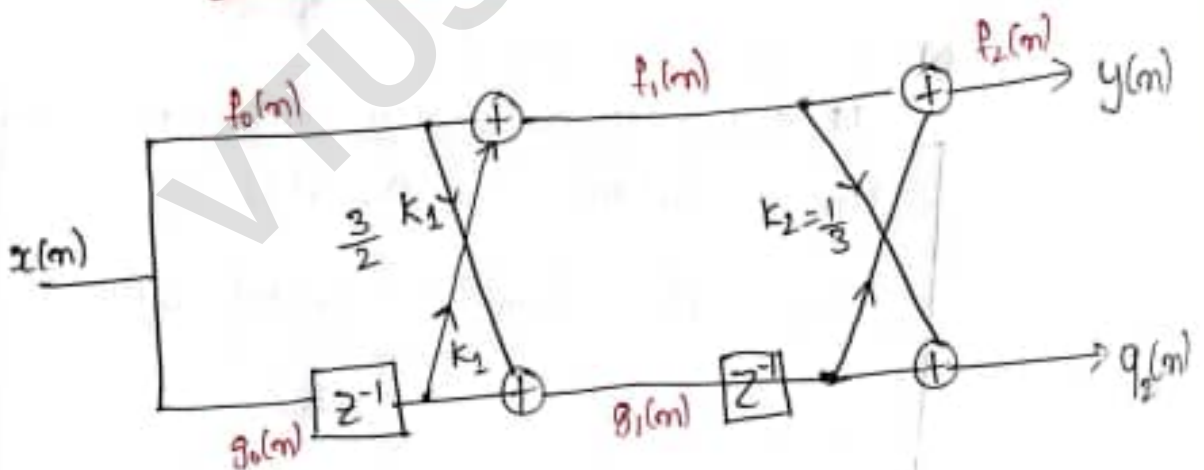
$$a_1(1) = \frac{a_2(1) - a_2(2) a_2(1)}{1 - k_2^2}$$

$$= \frac{a_2(1) [1 - a_2(2)]}{1 - \underbrace{(k_2^2)}_{a_2(2)}}$$

$$= \frac{a_2(1) [1 - \cancel{a_2(2)}]}{[1 - \cancel{a_2(2)}] [1 + a_2(2)]}$$

$$= \frac{a_2(1)}{1 + a_2(2)} = \frac{2}{1 + \frac{1}{3}} = \frac{2}{4/3} = \frac{6}{4} = \frac{3}{2}$$

$$k_1 = a_1(1) = 3/2$$



An FIR filter is given by  $y(n) = x(n) +$

$$\frac{2}{5} x(n-1) + \frac{3}{4} x(n-2) + \frac{1}{3} x(n-3)$$

Draw the lattice structure.

Soln:-  $y(n) = x(n) + \frac{2}{5} x(n-1) + \frac{3}{4} x(n-2) + \frac{1}{3} x(n-3)$

Taking z-transform,

$$Y(z) = X(z) + \frac{2}{5} z^{-1} X(z) + \frac{3}{4} z^{-2} X(z) + \frac{1}{3} z^{-3} X(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = 1 + \frac{2}{5} z^{-1} + \frac{3}{4} z^{-2} + \frac{1}{3} z^{-3}$$

Here  $a_3(1) = 2/5$ ,  $a_3(2) = 3/4$ ,  $a_3(3) = 1/3$ .

The filter order is 3.  $M=3$  i.e.,  $m=3, 2, 1$

W.K.T

$$K_m = a_m(m), \quad a_{m-1}(0) = 1$$

$$\Delta \quad a_{m-1}(i) = \frac{a_m(i) - a_m(m) a_m(m-i)}{1 - K_m^2} \quad i = 1, 2, \dots, m-1$$

STEP 1:

for  $m=3$ ,

$$\boxed{K_3 = a_3(3) = 1/3}$$

$$a_2(i) = \frac{a_3(i) - a_3(3) a_3(3-i)}{1 - K_3^2} \quad i = 1, 2$$

$$\therefore a_2(1) = \frac{a_3(1) - a_3(3) a_3(2)}{1 - K_3^2}$$

$$\begin{aligned}
 a_2(1) &= \frac{\frac{2}{5} - \frac{1}{3} \times \frac{3}{4}}{1 - \left(\frac{1}{3}\right)^2} = \frac{\frac{2}{5} - \frac{1}{4}}{1 - \frac{1}{9}} \\
 &= \frac{\frac{8-5}{20}}{8/9} = \frac{3}{20} \times \frac{9}{8} = \frac{27}{160} \\
 &= \underline{\underline{0.169}}
 \end{aligned}$$

$$\begin{aligned}
 \Delta \quad a_2(2) &= \frac{a_3(2) - a_3(3) a_3(1)}{1 - k_3^2} \\
 &= \frac{\frac{3}{4} - \frac{1}{3} \times \frac{2}{5}}{1 - \left(\frac{1}{3}\right)^2} = \frac{\frac{3}{4} - \frac{2}{15}}{8/9} \\
 &= \frac{\frac{45-8}{60}}{8/9} = \frac{37}{60} \times \frac{9}{8} = \frac{111}{160} \\
 &= \underline{\underline{0.693}}
 \end{aligned}$$

STEP 2:-  $m=2,$

$$\boxed{k_2 = a_2(2) = 0.693}$$

$$a_1(i) = \frac{a_2(i) - a_2(2) \cdot a_2(2-i)}{1 - k_2^2}, \quad i=1$$

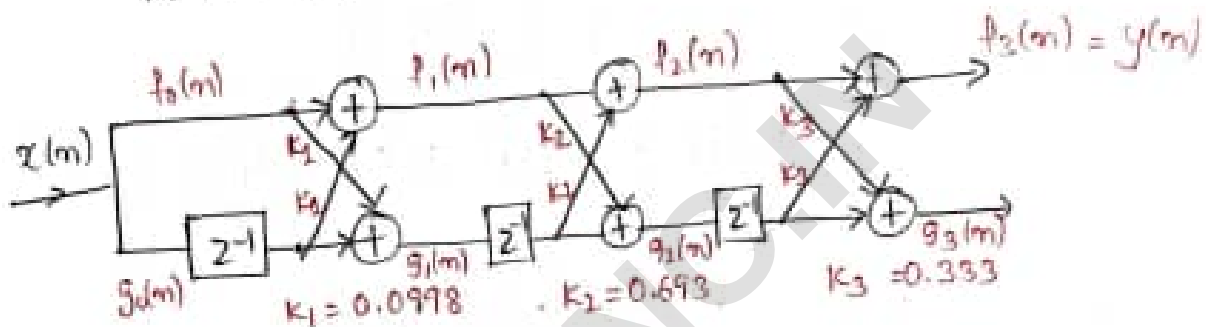
$$a_1(1) = \frac{a_2(1) - a_2(2) a_2(1)}{1 - k_2^2}$$

$$a_1(1) = \frac{\frac{27}{160} - 0.693 \times \frac{27}{160}}{1 - (0.693)^2} = \underline{\underline{0.0998}}$$

STEP 3:-  $m=1$

$$k_1 = a_1(1) = 0.0998$$

Hence the lattice structure is drawn.



Consider an FIR lattice filter with coefficients  $k_1 = 0.65$ ,  $k_2 = -0.34$ ,  $k_3 = 0.8$ . Find its impulse response. Draw the equivalent DF-I structure.

Soln:-

For  $m = 1, 2, \dots, M$ .

Consider  $a_m(0) = 1$

$a_m(m) = k_m$

$$a_m(i) = a_{m-1}(i) + a_m(m) a_{m-1}(m-i)$$

$$1 \leq i \leq m-1$$

Here  $m=3$ . i.e., 3 stages in the lattice structure

for  $m=1$ ,

$$a_1(0) = 1$$

$$a_1(1) = k_1 = 0.65$$

for  $m=2$ ,

$$a_2(0) = 1$$

$$a_2(2) = k_2 = -0.34$$

$$a_2(i) = a_1(i) + a_2(2) a_1(2-i), \quad i=1$$

$$\text{i.e. } a_2(1) = a_1(1) + a_2(2) a_1(1)$$

$$= 0.65 - 0.34 \times 0.65$$

$$a_2(1) = 0.429$$

for  $m=3$ ,

$$a_3(0) = 1$$

$$a_3(3) = k_3 = 0.8$$

$$a_3(i) = a_2(i) + a_3(3) a_2(3-i), \quad i=1, 2$$

$$i=1, \quad a_3(1) = a_2(1) + a_3(3) a_2(2)$$

$$= 0.429 + 0.8 \times (-0.34)$$

$$= \underline{\underline{0.157}}$$

$$i=2, \quad a_3(2) = a_2(2) + a_3(3) a_2(1)$$

$$= -0.34 + 0.8 \times (0.429)$$

$$= \underline{\underline{0.0032}}$$

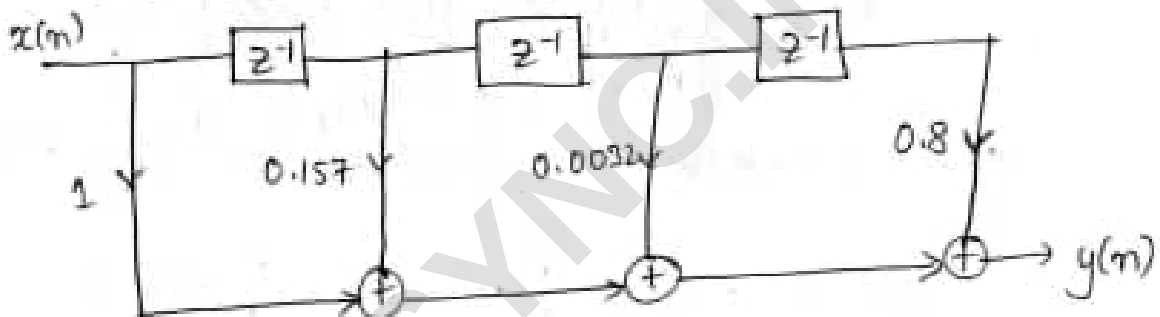
$$H(z) = 1 + a_3(1)z^{-1} + a_3(2)z^{-2} + a_3(3)z^{-3}$$

$$\therefore H(z) = 1 + 0.157z^{-1} + 0.0032z^{-2} + 0.8z^{-3}$$

Taking IZT,

$$h(m) = \delta(m) + 0.157\delta(m-1) + 0.0032\delta(m-2) + 0.8\delta(m-3)$$

DF-I Realization



Realization of Linear-Phase FIR filters:-

$$h(m) = h(m-1-m) \quad [\text{H(n) symm. about its origin}]$$

- ① Realize the linear-phase FIR filter having the following impulse response,

$$h(m) = \delta(m) + \frac{1}{4}\delta(m-1) - \frac{1}{8}\delta(m-2) + \frac{1}{4}\delta(m-3) + \delta(m-4)$$

Soln<sup>t</sup>

$$\text{Given } h(m) = \delta(m) + \frac{1}{4}\delta(m-1)$$

$$- \frac{1}{8}\delta(m-2) + \frac{1}{4}\delta(m-3) + \delta(m-4)$$

Taking z-transform on both sides & letting

$$H(z) = \frac{y(z)}{x(z)}, \text{ we get}$$

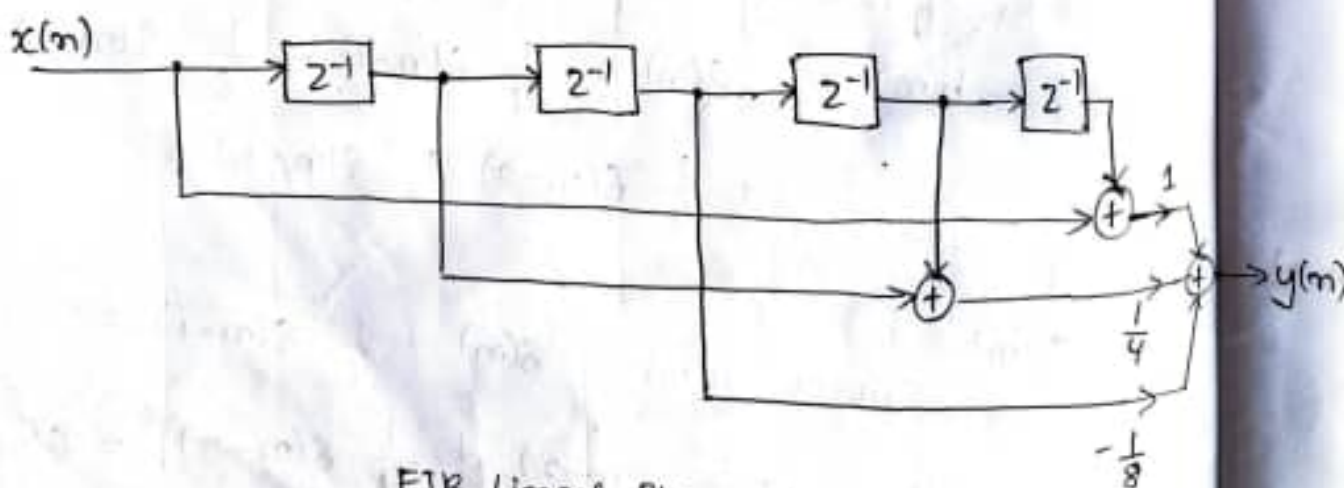
$$H(z) = \frac{y(z)}{x(z)} = 1 + \frac{1}{4} z^{-1} - \frac{1}{8} z^{-2} + \frac{1}{4} z^{-3} + z^{-4}$$

$$\Rightarrow y(z) = x(z) + \frac{1}{4} z^{-1} x(z) - \frac{1}{8} z^{-2} x(z) + \frac{1}{4} z^{-3} x(z) + z^{-4} x(z)$$

Taking inverse z-transform on both sides,

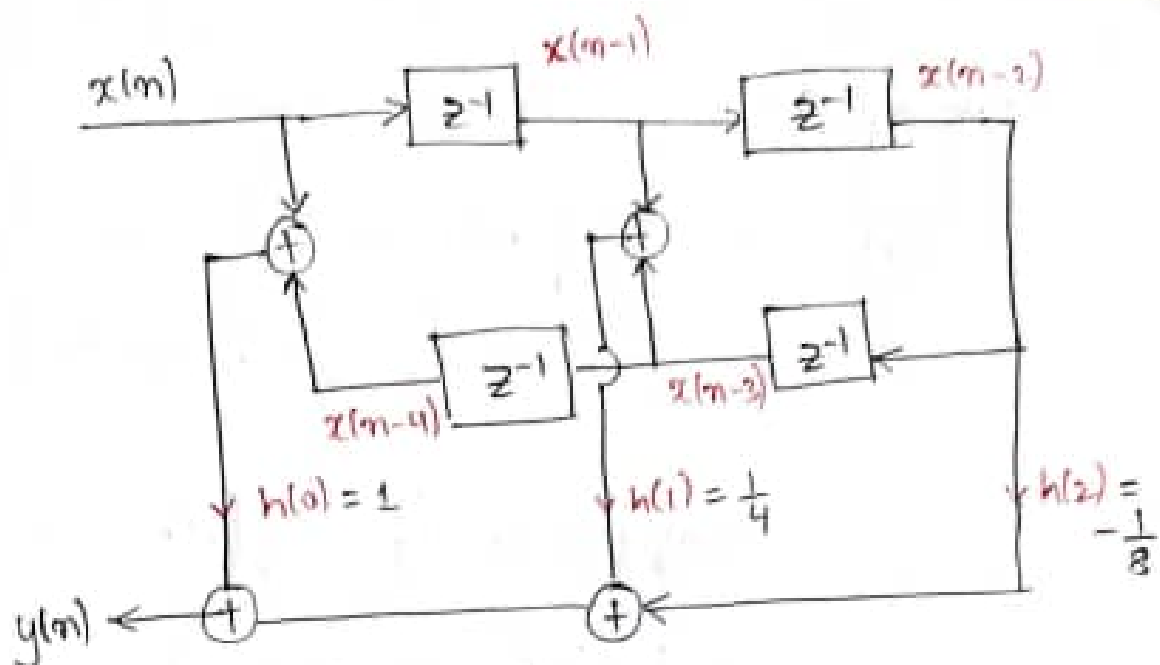
$$y(n) = x(n) + \frac{1}{4} x(n-1) - \frac{1}{8} x(n-2) + \frac{1}{4} x(n-3) + x(n-4)$$

$$= [x(n) + x(n-4)] + \frac{1}{4} [x(n-1) + x(n-3)] - \frac{1}{8} x(n-2)$$



FIR Linear Phase Realization





Realize the linear-phase FIR filter having the following impulse response.

$$h[n] = \delta[n] - \frac{1}{4} \delta[n-1] + \frac{1}{2} \delta[n-2] + \frac{1}{2} \delta[n-3] - \frac{1}{4} \delta[n-4] + \delta[n-5]$$

Soln: - Taking Z.T. we get.

$$H(z) = 1 - \frac{1}{4} z^{-1} + \frac{1}{2} z^{-2} + \frac{1}{2} z^{-3} - \frac{1}{4} z^{-4} + z^{-5}$$

$$\therefore \frac{Y(z)}{X(z)} =$$

$$\Rightarrow Y(z) = X(z) \left[ 1 - \frac{1}{4} z^{-1} + \frac{1}{2} z^{-2} + \frac{1}{2} z^{-3} - \frac{1}{4} z^{-4} + z^{-5} \right]$$

Taking IZT,

$$\begin{aligned}
 y(n) &= x(n) - \frac{1}{4} x(n-1) + \frac{1}{2} x(n-2) \\
 &\quad + \frac{1}{2} x(n-3) - \frac{1}{4} x(n-4) + x(n-5) \\
 &= [x(n) + x(n-5)] - \frac{1}{4} [x(n-1) + x(n-4)] \\
 &\quad + \frac{1}{2} [x(n-2) + x(n-3)]
 \end{aligned}$$

